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SCATTER GATHER (SG,44) 10/2/73 JBL/JGC

/FLAG 1=1_BLOCK,0_ITEM

/FLAG 2=1_HELD, 0_NOT HELD

/FLAG 3=1_NON-ADDRESSED, 0_ADDRESSED

```
ESG:      JSP .FSET
          DIO TRACK
          LAC B5
          SZF 2
          LAC B3
          SZF I 3
          CLA
          DIP ESG1
          LAW 1
          SZF I 3
          JMP ESG6
          SZF 1
          JMP .ESG5
          LAC I (DCH 70)
          DAP ESG4      /AC 6-17 NOW IN ESG4
          LAW I 7777
          AND I ESG4    /FIRST WORD OF FIRST COMMAND PAIR
          SAS B3        /WRITE
          JMP I 16SPCERR
          LAC I ESG4
          DAP ESG4      /ADDRESS OF FIRST WORD IN ITEM, I.E. WORD COUNT
          LAC I ESG4
          SAS ONE
          SPQ
          JMP I 16SPCERR
          MUL ONE      /TO DIVIDE FORMAT
          DIV C61
C7777,    7777
          SNI I
          IDA
.ESG5,    IOR TRACK

.ESG5+1,  JDA TRACK
ESG1,     000000

          JSP WAIT4
          JMP ESG3
```

/ADDRESSED SG OPERATION--CALL WHERE ONLY IF WRITING OR ZEROING

```

ESG6,    LAC I (DCH 70)
          DAP ESG4
ESG7,    LAC I ESG4      /GET COMMAND
          SPA
          JMP PSWHER     /DONE, NO WRITING
          SUB (040000)
          SMA
          JMP ESG2       /FOUND A WRITE, CALL WHERE
          IDX ESG4
          IDX ESG4       /NEXT COMMAND
          JMP ESG7

```

```

PSWHER,  LAC I (DCH 72) /WHERE DOES THIS FOR TRACK
          SPA
          JMP I 16SPCERR
          SAR 1S
          IAI
          JMP .ESG5+1

```

```

ESG2,    JSP WHERE
          JMP .ESG5+1

```

```

/FLAG 4=1 FIRST BLOCK CHECKING ALREADY DONE
/FLAG 5=1 WRITING ALLOWED IN THIS ITEM

```

BUFR=170000 BUFF

```

ESG4,    0
ESG3,    CLF 6
ESG5,    CLF 4
          CLF 5
          SZF 2
          STF 5
          CLF 2
          LAW 7777
          AND I AC
          IOR B3
          DAC TABPTR
          LAC I IO
          DAC NXTTAG
          LAW CMPERR
          DAP COMPX
          DZM RWSTAT
          DZM ERSTPS
          DZM COMSTP
          DZM COMPST
          DZM BLKSUM
          LAW BUFF=1
          DAP GBKADR
          SZF I 1
          JMP SGIT

```

/READ SCATTER GATHER BLOCK

SGBL,	STF 5	/ALLOWS WRITES
	STF 4	/DON'T DO FIRST BLOCK OF ITEM JAZZ
	JSP GTCMTA	
	JSP SVPTRS	
	LAW 50.	
	DAC LNTH	
	LAC NXTTAG	
	SZF 3	
	JSP GBKAD	
	DAC NXTTAG	
	DAC TAGWD	
	LAW 100	
	SZF I 3	
	DAC COMPST	
	LAC THSBUF	
	DAP SGBL1	
	DAC CSPTR	
	JSP SGCS	
	LAC NXTTAG	
	JDA .READ	
	JSP GO	
SGBL1,	I I I .	
	LAC BUFF	
	JMP SGBLX	

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SGTAP1, JSP GO
SGTAP2, I I I
SGTAP3, JMP SWAPE1

SGIT, LAW 49.
DAC LNGTH
DZM CMDWCT
JSP GTCMTA /GET FIRST TABLE PAIR
DZM TOTBLK
DZM PULPTR /?
DZM BLKSWD
DZM 1ERR

LAC NXTTAG
SZF 3
JSP GBKAD
DAC TAGWD
DAC NXTTAG+2
SZF I 3
JMP IT03
LAC I OWNWRD
DAC NXTTAG+1
STF 5
JSP GBKAD
SAD MZERO
LAW 1
IOR B0
DAC NXTTAG
LAC B3
IOR WD1
DAC PULPTR
JMP IT03

/GET THE WORD COUNT

GBKADI, DAP GBKADX
JMP GBKADX+4
GBKAD, DAP GBKADX
IDX .+1
GBKADR, LAC .
SMA
GBKADX, JMP .
LIO FBLOCK
SNI
JMP GBKADX
GBKADX+4, LAW BUFF
DAP GBKADR
JSP WAIT
LAC TAGWD
JDA GBLOCK
JMP GBKADR

IT0, SZF I 3
 JMP IT03
 JSP GBKAD
 SPA
 LAW 1
 DAC NXTTAG

IT03, DZM RWSTAT
 LIO THSBUF
 LAC NXTBUF
 DIO NXTBUF
 DAC THSBUF
 DAC CSPTR
 DAP IT2
 JSP SVPTRS
 JSP SGCS

IT1, SZF 4
 JSP WAIT
 LAC TAGWD
 SAD ONE
 JSP ILLITM
 JDA ,READ
 JSP GO

IT2, I I I .

```

IT6,      LAC NXTTAG      /JUST READ OR WRITTEN, IN EITHER
          DAC TAGWD       /CASE, THE NEXT BLOCK TO BE ACCESSED
          SZM"U"SZF 4
          JMP SGIT8       /NOT FIRST BLOCK AND AC>0
          SPA
          SZF 4
          JSP WILLIT
          XOR B0
          SAD NXTTAG+2
          JSP WILLIT
          CLA"U"STF 4     /FIRST BLOCK AND AC<0
          LIO I PULPTR
          SAS PULPTR
          DIO WRDCNT
          LAC NXTTAG+2
          DAC NXTTAG+1
          LAC NXTTAG+1
          SAS NXTTAG+2
          JSP WILLIT
          CLA
          SAS ERSTPS
          STF 2           /WORKS

          SZF I 2
          JMP .+3
          SAD NEWDCI      /QUIT IF NO CHANGE IN LENGTH OF ITEM
          JMP SGIT20      /AND NO MORE COMMANDS
          LAW 49.
          ADD TOTBLK
          DAC TOTBLK
          SUB WRDCNT
          SMA
          JMP SGIT20      /LAST BLOCK ALREADY DONE
          ADD C61
          DZM BLKSUM
          SPA
          JMP IT0
          DAC BLKSUM

```

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```
SGIT11,  LAC BLKSWD      /LAST BLOCK IN CURRENT ITEM COMING NEXT
          SZA I
          JMP IT0         /NO CHANGE IN THE NO. OF BLOCKS
          IDX RWFLAG
          STF 3
          JMP RSTRT1
```

```
RBIBUF,  DAP RBIBFX      /READ CURRENT BLOCK INTO BUFF 100
          LAC TAGWD
          DAP DCS+1
          JDA ,READ
          JSP GO
          I I I DCS
RBIBFX,  JMP .
```

```
DCS,     560001
          200000 .
CMUMBL,  630000 NXTTAG
          040001
DCS+4,   630000 BUFF 100
          100062
          000000
```

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SGIT20,

JSP WAIT
CLA
SAS 1ERR
JMP 1ERR+1
LAW I 7777
AND I TABPTR
SAS (770000
JSP ILLSPC
LAC WRDCNT
SUB CMDWCT
SPA
JSP ILLSPC
LAC NXTTAG+2

/WAY OUT

/NO. OF WORDS REQUESTED
/TO BE PROCESSED
/GREATER
/THAN WORD COUNT

SGBLX,

SZF 3
DAC I IO
JMP SWAPE1

SGRET1,

/BEGINNING OF THE ACTUAL DC SEQUENCE SETTING UP

SGCS,	DAP REDX	
	LAC RITC1	/(560101
	SZF I 3	/FLAG 3=1, NON-ADDRESSED, 0, ADDRESSED
	SZF I 1	/FLAG 1=1, BLOCK, 0, ITEM
	XOR C100	/NOT AN ADDRESSED BLOCK
	DAC I CSPTR	
	IDX CSPTR	
	LAC .	/READ TAG, READ DATA
	SZF 3	
	LAC SGCS	/WRITE TAG, WRITE DATA
	DIP I CSPTR	
	LAC TAGWD	
	SZA I	
	JSP WILLIT	
	DAP I CSPTR	
	IDX CSPTR	
	LAC CMUMBL	/(630000 NXTTAG
	DAC I CSPTR	
	IDX CSPTR	
	LAC C60001	
	SZF 1	
	JMP SGCS1	/BLOCK
	LAC C50001	/ITEM
	DAC I CSPTR	
	IDX CSPTR	
SGCS1,	LAC C40001	
	SZF 3	
	AND (-20000	/NON-ADDRESSED BLOCK
	DAC I CSPTR	
	IDX CSPTR	
	SZF I 4	/FIRST BLOCK OF ITEM SWITCH
	JMP FBOI	

DISPAT, LAW DISPTB
 ADD COMMAN
 DAP DISPT1
 SZF I 2
DISPT1, JMP .

EOTAB, LAW I 1
 ADD CSPTR
 DAC CSPTR
 LAC I CSPTR
 AND (37777)
 JMP RED11

/CHANGE 0400XX TO 1000XX

DISPTB, JMP IREAD
 JMP SKIP
 JMP COMPAR
 JMP NONCOM
 JMP WRIT
 JMP WZERO

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```
RSTATC,  DAP RSTATX  /READ MODE STATUS CHECK
          LAC RWSTAT  /IO TRANSPARENT
          SAD ONE
RSTATX,  JMP .
          SZF 3
          JMP REGBOU
          SZA
          JMP RESTRT
          IDX RWSTAT
          JMP RSTATX

SKIP,    JSP RSTATC
          LAC DCS+4    /UNTIL DAC ALL 1'S IS RELIABLE
          JMP RED1

CLIO,COMP,  LIO C100
          JMP . 2

NONCOM,    LIO C200
          JSP RSTATC
          LAC COMPST
          IAI
          XAI
          SZA
          JMP RESTRT  /BOTH REQUESTED IN SAME BLOCK
          DIO COMPST
          LAC I THSBUF /START OF COMMAND SEQ. = 560001
          IAI
          DAC I THSBUF
          LAC B4       /DATA DIRECTION BIT
          DAC COMBPT
          JMP RED05    /SKIP ERROR CHECKING
```

FORM 3510
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IREAD,	JSP RSTATC	
RED00,	LAC B5	/(I)
	SUB WD1	
	SUB WD2	
	SPA	
	JMP REGMAX	
RED04,	LAC WD1	
	SUB BOUND	
	SPA	
	JMP REGBOU	/BOUND ERROR
RED05,	LAC WD1	
	IOR SCORE	
RED1,	DAC I CSPTR	/DATA ADDRESS
	IDX CSPTR	
	LAC BLKSUM	
	SUB LENGTH	
	LIA	
	ADD WD2	
	SPQ	
	JMP RED2	
	DAC WD2	
	LAC LENGTH	
	DAC BLKSUM	
	CMI"U"LAI	
	ADD WD1	
	DAC WD1	
	LAI	
RED12,	IOR COMBPT	
RED11,	IOR B2	
	DAC I CSPTR	/WD COUNT
	IDX CSPTR	
RED13,	DZM I CSPTR	/D.C. HALT
	LAW 2	
	SAS RWSTAT	
REDX,	JMP .	
	LAC LENGTH	
	SAS BLKSUM	
	SZF 3	
	JMP REDX	
	JMP RESTRT	

RED2,

LIA
ADD LENGTH
DAC BLKSUM
LAC B3
SNI
LAC B2
IOR WD2
IOR COMPBT
DAC I CSPTR
CLA
SAD COMSTP
JMP .+4
IDX ERSTPS
LAC WD2
JMP RED12
IDX CSPTR
SNI I
JMP GTCMTB
JSP GTCMTA
JMP RED13

```
WSTATC,   DAP WSTATX      /WRITE MODE STATUS CHECK
          SZF I 5
          JMP REGBOU
          LAC RWSTAT
          SAD TWO
WSTATX,   JMP .
          SZA
          JMP RESTRT
          IDX RWSTAT
          ADD THSBUF
          DAC JUNK
          LAC B4
          IOR I JUNK
          DAC I JUNK
          IDX RWSTAT
          JMP WSTATX

WZERO,    JSP WSTATC
          CLC
          JMP RED1

WRIT,     JSP WSTATC
          JMP RED05

REGMAX,   ADD WD2
          DAC WD2
          LAC B2
          DAC 1ERR
          DAC COMSTP
          JMP RED04

REGBOU,   LAC B3
          DAC 1ERR
          STF 2
          IDX ERSTPS
          JMP EOTAB
```

/FIRST BLOCK OF ITEM ROUTINE
/DATA CHANNEL COMMANDS ALREADY SET UP:

/ 560001
/ 20XXXX
/ 630000+NXTTAG
/ 050001
/ 040001

FB01,	SZF 3	
	JMP DISPAT	
	SZF 2	
	JMP SGRET1	
	DZM WRDCNT	
	DZM NEWDCNT	
	LAC COMMAN	
	SZA	/READ
	SAD TWO	/CONT. ON COMPARE
	JMP FB11	/SET UP PULPTR
	SAS THREE	/CONT. ON NON-COMPARE
	SAD FOUR	/WRITE
	JMP FB02	
	SAD ONE	/SKIP
	JMP FB10	/READ ONE WORD INTO WRDCNT
	JSP ILLSPC	
FB039,	DAP FB0399	
	IDX WD1	/SUBROUTINE USED TWICE, SAVES 2 REGISTERS
	LAW I 1	
	ADD WD2	
	DAC WD2	
	SZA I	
	JSP GTCMTA	
FB0399,	JMP .	
DIVC61,	DAP DVC61X	
	RIL 18	
	CLA	
	DIV C61	
C777,	777	
	SNI I	
	IDA	
	SAD ONE	
	STF 3	
DVC61X,	JMP .	

/FOR SKIP THE WORD COUNT
 FB10, LAC (630000+WRDCNT

DAC I CSPTR

IDX CSPTR

LAC C40001

DAC I CSPTR

IDX CSPTR

JSP FB039

IDX BLKSUM

LAW 4

SAS COMMAN

JMP DISPAT

/CONTINUE WITH SGCS

FB01, JSP UNSVPT

JMP FB02

FB11, LAC WD1

/SET UP PULPTR, TO TRANSFER ONE WORD FROM USER CORE

IOR B3

/ON READ OR CONT, ON COMPARE OF WORD COUNT

DAC PULPTR

LAW 1

SAS WD2

JMP DISPAT

LAW I 7777

AND I TABPTR

SAS B3

JMP DISPAT

FB02, JSP RBIBUF

/READ BLOCK INTO BUFF

LAC NXTTAG

SMA

JSP ILLITM

LAC BUFF 101

DAC WRDCNT

SZM

SAD ONE

JSP ILLITM

LAC WD1

IOR B3

DAC SPPTR1

LAW 4
 SAS COMMAN
 JMP FB03
 LAC I PTR1
 SAD WRDCNT
 JMP FB03
 DAC NEWDCT
 SZM
 SAD ONE
 JSP ILLSPC
 LIA
 JSP DIVC61
 DAC BLKSWD
 LIO WRDCNT
 JSP DIVC61
 SUB BLKSWD
 DAC BLKSWD
 DAC FBLOCK
 SZA I
 CLF 3
 LIO NEWDCT
 SMA
 DIO WRDCNT

FB03, JSP FB039
 LAC WD1
 IOR B3
 DAC SPPTR1
 LAC COMMAN
 SAD (5)
 JSP ILLSPC
 SAS FOUR
 JMP FB08

FB07, SZF 5
 JMP FB09
 LAC BUFF 100
 DAC NXTTAG+1
 SAS I OWNWRD
 JMP RTERR
 LAC I SPPTR1
 SAS BUFF 102
 JMP RTERR
 IDX I SPPTR1
 FB09, LAC I SPPTR1
 DAC BUFF 102

STF 5
 FB08, IDX ALREDS

/IF ENTIRE FIRST BLOCK IS TO BE WRITTEN

/PASS WORD INCORRECT

/REWRITE NUMBER ERROR

/INCREMENT REWRITE NO.S

/INDICATES WRITING LEGAL IN THIS ITEM

/SET ALREADY READ SWITCH

/HERE WHEN INCOMPATIBLE COMMANDS OCCUR

RESTRT,	JSP UNSVPT DZM RWFLAG	/RESTORE POINTERS TO START OF BLOCK STATE
RSTRT1,	DZM BLKSUM LAC ALREDS DZM ALREDS SZA JMP REST15 JSP WAIT JSP RBIBUF	/ENTRY FOR TRANSITION BLOCK /CHECK IF ALREADY READ /IF SO, SKIP READING AND ERROR CHECKING
REST0,	SZF 1 JMP REST1 LAC NXTTAG+2 SZF 4 SAD BUFF 100 JMP REST15 JSP ILLITM	
REST1,	LAC NXTTAG SAS TAGWD JSP ILLITM	/NOT A BLOCK
REST15,	LAC (BUFR 100) SZF I 1 IDA DAC SPPTR2	

```
REST2,  LAC SPPTR2
        SAD (BUFR 162
        JMP FINA
        LAC WD2
        SZA I
        JSP GTCMTA
        SZF 2
        JMP FINA
        LAC WD1
        IOR B0+3      /RENAME IOP USER CORE
        DAC SPPTR1
        LAC WD2
        ADD BLKSUM
        SUB LENGTH
        SZM
        JMP REST3
        DZM WD2
        ADD LENGTH
        DAC BLKSUM
        ADD (BUFR 100)
        SZF I 1
        IDA
REST3,  JMP REST4
        DAC WD2
        LAC WD1
        ADD LENGTH
        SUB BLKSUM
        DAC WD1
REST4,  LAC (BUFR 162
        DAC TERMIT
        LAW DISP2
        ADD COMMAN
        DAP .+2
        SZF I 2
        JMP .
        JMP FINA

DISP2,  JMP RREAD
        JMP RSKIP
        JMP RCOMP
        JMP RNONC
        JMP RWRIT
        JMP RZERO
```

RREAD, LAW 7777
 AND SPPTR1
 SUB BOUND
 SPA
 JMP INTBOU /BOUND ERROR

RREAD1, LAC (407777)
 SUB SPPTR1
 SPA
 JMP INTMAX
 LAC I SPPTR2
 DAC I SPPTR1
 IDX SPPTR1
 IDX SPPTR2
 SAS TERMIT
 JMP RREAD1
 JMP REST2

RSKIP, LAC TERMIT
 DAC SPPTR2
 JMP REST2

RWRIT, SZF I 5
 JSP ILLSPC
 IDX RWFLAG

RWRIT3, LAC I SPPTR1
 DAC I SPPTR2

RWRIT4, IDX SPPTR1
 IDX SPPTR2
 SAS TERMIT
 JMP RWRIT3
 JMP REST2

```

RCOMP,    LAW LOSE
          DAP COMP1
          LAW REST2
          DAP COMP2
          JMP COMPS

RNONC,    LAW RSKIP      /WIN
          DAP COMP1
          LAW LOSE
          DAP COMP2

COMPS,    LAC I SPPTR1
          SAS I SPPTR2
COMP1,    JMP .
          IDX SPPTR1
          IDX SPPTR2
          SAS TERMIT
          JMP COMPS
COMP2,    JMP .

RZERO,    SZF I 5
          JSP ILLSPC
          IDX RWFLAG
          DZM I SPPTR2
          IDX SPPTR2
          SAS TERMIT
          JMP .-3
          JMP REST2

CMPERR,   RRI 700      /GET DAC
          LAW I 1
          AAI
          AND (177777)
          SAD (I I I NXTTAG)
          JSP ILLITM
          AND C7777
          LIA
          LAW 3      /CONT. ON NON-COMPARE MUST BE DONE INTERPRETIVELY
          SAS COMMAN
          JMP CMPER2
          CLA
          SAS WD2      /THIS IS A CLUDGE
          JMP IT6
CMPER2,   DIO I ERCOD2
CMPER1,   LAW 4000
          DAC 1ERR
          CLA
          SZF I 1
          SAD NEWDCT
          JMP 1ERR+1
          STF 2
          JMP IT6

```

FINA,	LAW FINX+1
FINB,	DAP FINX
	CLA
	SAD WD2
	JSP GTCMTA
	CLA
	SAD RWFLAG
	JMP FINX
	SZF I 3
	JMP FINE
	LIO BLKSWD
	SPI
	JMP FINZ
	LAC NXTTAG
	DAC DRMPTR
	LAW EXP4
	LAW ILLITM
	DAP COMPM
	LAC I IO
	DAC PTR1
	LAW 1
	JMP FINC
FINZ,	LAC NEWDCT
	DAC WRDCNT
	DZM BLKSWD
	JSP GBKADI
FINC,	SZF I 4
	IOR B0
	DAC NXTTAG
	LAC B3
FINE,	IOR CLIO
	DIP DCS+1
	LAC TAGWD
	JDA ,READ
	JSP GO
	I I I DCS
	LAC .
	DIP DCS+1
FINX,	JMP .
	SZF I 1
	JMP IT6
	JMP SWAPE1
LOSE,	JSP FINB
	LAW 7777
	AND SPPTR1
	DAC I ERCOD2
	JMP CMPER1
INTMAX,	LAC B2
	JMP .+2
INTBOU,	LAC B3
	DAC 1ERR
	STF 2
	JMP FINA

/GET NEXT COMMAND PAIR

GTCMTB,	LAW DISPAT	/OPEN ROUTINE EXIT
GTCMTA,	DAP GTCMTX	/CLOSED ROUTINE ENTRY POINT
	DZM COMBPT	
	DZM WD1	
GETCM3,	LAC I TABPTR	/OR SOME SUCH
	DAP WD1	
	RAL 6S	
	AND (77)	
	DAC COMMAN	
	SAD (77	
	JMP GETCM4	
	SUB (6	
	SMA	
	JMP GETCM5	
	IDX TABPTR	
	LIO I TABPTR	
	DIO WD2	
	LAI	
	ADD CMDWCT	
	DAC CMDWCT	
	IDX TABPTR	
	SNI	
	JMP GETCM3	
	SPI I	
GTCMTX,	JMP .	
GETCM5,	LAC B3	
	DAC 1ERR	
GETCM4,	STF 2	
	JMP GTCMTX	
SVPTRS,	DAP SVPTRX	
	LAC WD1	
	DAC SWD1	
	LAC WD2	
	DAC SWD2	
	LAC COMMAN	
	DAC SCOMMA	
	LAC TABPTR	
	DAC STABPTR	
	LAC CMDWCT	
	DAC SCMDWC	
SVPTRX,	JMP .	
UNSVPT,	DAP UNSVPX	
	LAC SWD1	
	DAC WD1	
	LAC SWD2	
	DAC WD2	
	LAC SCOMMA	
	DAC COMMAN	
	LAC STABPTR	
	DAC TABPTR	
	LAC SCMDWC	
	DAC CMDWCT	
	CLF 2	
UNSVPX,	JMP .	

/VARIABLES

THSBUF, BUFR 162
 NXTBUF, BUFR 331

TAGWD, 0
 NXTTAG, 0

COMPST, 0
 COMPBT, 0

WD1, 0
 WD2, 0
 COMMAN, 0
 CMDWCT, 0

TOTBLK, 0
 BLKSWD, 0

/TOTAL NO. OF WORDS PROCESSED SO FAR THIS ITEM
 /3 WAY SWITCH RE LENGTH OF ITEM
 /0 = NO CHANGE
 /+ = GETTING SHORTER
 /- = GETTING LONGER

RWSTAT, 0
 WRDCNT, 0

/READ-WRITE STATUS
 /NO. OF WORDS IN ITEM NOW, EXCEPT AFTER ADDING TO IT
 /STARTS, WHEN IT IS NO. OF WORDS IN ITEM
 /WHAT WRDCNT WILL BE AFTER ADDING STARTS BEFORE

NEWDC, 0
 JUNK, 0
 RWFLAG, 0
 ALREDS, 0
 TERMIT, 0
 ERSTPS, 0
 COMSTP, 0
 SWD1, 0
 SWD2, 0
 SCOMMA, 0
 STABPTR, 0
 SCMDWC, 0

/A TEMP STORAGE REG
 /SWITCH FOR INTERP, SIGNALS THAT THE BLOCK IS REWRITING
 /BLOCK ALREADY READ SWITCH
 /1> THAN ADDR OF LAST WORD IN BUFF+100 PROCESSED
 /ERROR STOP SWITCH, AFTER COMMANDS EXECUTED
 /ERROR STOP SWITCH, AFTER THIS COMMAND SET UP

TABPTR, 0
 BLKSUM, 0
 LGTH, 0

/49. OR 50.

CSPTR, 0
 TAPCTR, 0

/COUNTS THE NUMBER OF COMMAND PAIRS USED

/FASTRAND ERROR RETURNS

```

1ERR,      0
            JSP WAIT4
            LAC 1ERR
            LIO DCONT      /IS IT TAPE
            SPI
            JMP I TERR
            LIO SWPFLG
            LFI
            DAC I ERCOD1
            LAW I 1
            SZF 6
            LAW I 2
            ADD I PC
            LIA
            DAC I TRAPPC
            IDA
            SZF I 6
            LAW IOPTSU
            SPI
            IOR B0
            DAC I PC
            JMP SWAPE1

```

FHLT,
1HLT,

```

JSP METH1
CLC
DAC ONTRAC
LAC B1
JDA 1ERR

```

/CHANNEL 4 TOTAL CRASH

/FASTRAND=CHANNEL 1 TOTAL CRASH

MAXERR,

```

LAC MAXM
SUB CORADR
IDA
DAC I ERCOD2
LAC B2
JDA 1ERR

```

/GIVE NO. OF WORDS READ TO USER

ILLSPC,

```

DAC I ERCOD2
LAC B3
JDA 1ERR

```

NSERR,

```

LAW 400
JDA 1ERR

```

/ERROR RETURNS WHEN THE CHANNEL STILL FLYING

```
WILLIT,  LIO B8
          DIO 1ERR
W1ERR,   DAC I ERCOD2
          JSP WAIT
          LIF
          RCR 68
          DIP I ERCOD2
          JMP 1ERR+1
```

/MOVE BIT TABLES OUT

DONE, DAC DONX
 DSC 100
 LSM
 CLI /START WITH 1ST BIT TABLE

DON5, DIO DON4
 JSP BTWRIT /SET UP COMMANDS
 JMP DON6 /NONE THERE
 LIO (30000 NDONC1)
 JSP DGO /WRITE IT OUT

DON6, CLC
 DAC ONTRACK
 IDX DON4
 LIA
 SAS (4) /DO ALL BIT TABLES

JMP DON5
 LAC (270000)
 DIP NDONC2+1
 LIO (30000 NDONC2)
 JSP DGO
 ASC 100
 ESM
 JMP I DONX

DONX, 0
 DON4, 0

DGO, DAP DGOX
 LAI
 DGO+2, RRO+500
 XX

RRI
 RIR 5S
 SPI
 JMP .-3
 RIL 5S
 SPI I

DGOX, JMP . /AGH ! (ABNORMAL)
 LIA
 JMP DGO+2

NDONC2, 560001
 213000 /FIXED-HEAD 3 HOLDS FRELST WHEN EXEC NOT RUNNING
 617000
 040677
 100001
 000000

```

NDONC1, 560001
        300000 .
        140000
        260000 .
        610000 .
        040100
        630000 .
        100001
        000000

```

/SUBROUTINE TO WRITE OUT BIT TABLE

```

BTWRIT, DAP BTWRX
        LAW TRACKA
        AAI
        /WHICH TABLE ? 3>=IO>=0
        DAP BTWR1
        DAP BTWR2
        LAW CNT1
        AAI
        DAP NDONC1+6
        LAW BITTB1
        RIL 6S
        AAI
        DAP NDONC1+4
BTWR1,  LAC .
        SPA
        /IS THERE A BIT TABLE HERE ?
        JMP BTWRX
        /NO. R1
        CLI"U"CM1
BTWR2,  DIO .
        /MARK IT
        SUB (FRELST)
        CLI
        RCR 2S
        IOR C400
        DAP NDONC1+1
        LAI
        RAR 6S
        DAP NDONC1+3
        IDX BTWRX
        /R2 AFTER SETTING UP COMMANDS
BTWRX,  JMP .

```

/CORE 4,0 REFERENCES FROM CORE 17

C16A,	CAL 0
C16B,	CAL 1
IOPMAX,	IOR 74
ERCODE1,	IOR 102
ERCODE2,	IOR 103
OWNWRD,	IOR 104
TRAPPC,	IOR 77
BOUND,	36
OVERFLOW,	=0
OVFX,	JMP .
COMPX,	JMP .

CONS, CONSTANTS

FOO,	FLEXO FOO
37200/	BUFF,
37700/	
CH1BRK,	DCH .
CH15BK,	DCH .
R0,	DCH .
R1,	DCH .
R2,	DCH .
SWORG,	DCH .
HOTFLG,	DCH .
RSTAT,	DCH .
MSTAT,	.
INIT,	DCH .
DSWAPX,	DCH .
DCORE,	DCH .

37720/	
LISTPC,	TAD .
ELSTPC,	TAD .
SWAP4,	TAD .

/NON CORE 16 STARTUP

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AC, JDA 7775

37724/
T1, XCT . /CORE 10 POINTER

PC, IOR 35
IO, JDA 7776
FLAGS, JDA 7777

37730/
ABNORMAL, DCH I .
CONTR2, DCH I .
FTAPE1, DCH I .
4CHAN1, DCH I .
NOTYOURS, DCH I .
16SPCERR, DCH I .
EMPTY, DCH I .
TERR, DCH I .
TAPPAR, DCH I .
TAPSTT, DCH I .
TUNIT, DCH I .
INDEX1, DCH I .
SWAPE7, DCH I .
SWAPE2, DCH I .
NETHND, DCH I .

START HLT-JMP